Section IV

INTRAVENOUS FLUID THERAPY

Life is water dancing to the tune of solids.

Albert Szent-Gyorgi

Chapter 10

Intravenous Fluids

The secret of science is to ask the right question …

Sir Henry Tizard (*a*)

The first intravenous fluid was a mixture of opium, wine and beer, which was infused through a goose quill into the vein of a dog. The year was 1656, and the infusion was carried out by a well-known London architect named Christopher Wren (the designer of St. Paul's Cathedral in London). The dog's response to the infusion was stated as follows: *"It made him extremely drunk, but soon after he Pisseth it out"* (1). This rather comical report received little attention, as the dominant approach to illness in the 17th and 18th centuries involved the *removal* of fluid by bloodletting and the use of purgatives. The therapeutic benefit of intravenous fluids was first recognized during the cholera epidemic in the early 19th century, when the infusion of sodium chloride solutions had some success in improving outcomes (2). Thereafter, intravenous fluids gained in popularity for the rehydration of patients with gastroenteritis, but it wasn't until the 1940s, with the advent of blood fractionation and the violence of World War II, that intravenous fluid therapy emerged as a staple of circulatory resuscitation.

Simply stated, the care of critically ill patients would not be possible without the use of asanguinous intravenous fluids, which are classified as crystalloid or colloid fluids based on their tendency to pass through capillary walls. This chapter describes the salient features of these fluids, both individually and as a group.

SOME BASICS

Body Water Distribution

The distribution of fluid in the adult human body is summarized in the following statements.

. Total body water (TBW) accounts for about 60% of the lean body weight (600 mL/kg) in adult males, and 50% of the lean body weight (500 mL/kg) in adult females.

. About two-thirds (≈65 %) of the TBW is located intracellularly, and the remaining one-third (≈35%) is in the extracellular fluid compartment.

. Plasma represents only one-fifth (20%) of the extracellular fluid, and the remaining 80% is located in the interstitial space (discounting the small volumes allotted to cerebrospinal fluid and pleural fluid).

The above information is included in Table 10.1, along with representative body fluid volumes for an average-sized adult male and female. Note the small volume of plasma, which represents only 6−7% of the total body fluid. This is relevant because the goal of intravenous fluid therapy is to support the plasma volume.

TABLE 10.1 Body Fluid Volumes in Adults

Parameter	Derivation	Male	Female
Lean Body Weight		75 kg*	60 kg*
Total Body Water	0.6 x LBW (M) 0.5 x LBW (F)	45 L	30 L
Intracellular Volume	0.65 x TBW	29 L	20 L
Extacellular Volume	0.35 x TBW	16 L	10 L
Interstitial Volume	0.8 x ECV	13 L	8 L
Plasma Volume	0.2 x ECV	3 L	2 L

* Representative weight for an average-sized adult. LBW = lean body weight, TBW = total body water, ECV = extracellular volume.

Osmotic Forces

(*Note:* For a more detailed description of this topic, see the first section of Chapter 35.)
The distribution of water in the intracellular and extracellular fluid compartments is determined by the relative osmotic activity in each compartment; i.e., the compartment with the higher osmotic activity will draw water from the one with the lower osmotic activity. (Relative osmotic activity is often expressed as *tonicity*.) Osmotic activity is determined by the number of solute particles in a solution, and not by the size, electrical charge, or chemical behavior of the solutes. *Sodium is the most abundant solute in the extracellular fluid, and is thus the principal determinant of extracellular fluid volume*. Osmotic activity can be expressed as *osmolarity* (osmotic activity per unit volume of solution) or *osmolality* (osmotic activity per unit volume of water). Extracellular fluid (plasma) is 93% water, so osmolarity and osmolality are often used interchangeably.

Colloid Osmotic Pressure

The distribution of water between the intravascular and extravascular compartment is influenced by the osmotic activity of large molecules in plasma that do not pass freely across the vascular endothelium. Plasma proteins are the principal source of this osmotic activity, which acts to hold water in the intravascular compartment. The osmotic force generated by large, semipermeable or impermeable molecules is called the *colloid osmotic pressure*, or the oncotic pressure. *Albumin is the most abundant plasma protein, and is the principal source of the colloid osmotic pressure*

in plasma.

CRYSTALLOID FLUIDS

Crystalloid fluids are aqueous solutions that contain small molecules (mostly electrolytes) that diffuse freely from intravascular to interstitial fluid compartments. The principal component of crystalloid fluids is sodium chloride.

Normal Saline

The most frequently used crystalloid fluid is 0.9% sodium chloride (0.9% saline), with annual sales of 200 *million* liters in the United States (data from Baxter Healthcare). The popular name for this solution is *normal saline*, which is attributed to a Dutch chemist named Hartog Hamburger, who studied the freezing point depression of plasma in the late 19th century, and concluded incorrectly that plasma is a 0.9% salt solution (3). (Human plasma is actually a 0.6% salt solution.) A more appropriate name for 0.9% saline is *isotonic saline*, but even this name is not accurate (see next).

Normal Saline is NOT Normal

Normal saline (0.9% saline) is not chemically normal because a one-normal (1 N) NaCL solution contains 58 grams of NaCL per liter (the combined molecular weights of sodium and chloride), while a 0.9% NaCL solution contains only 9 grams of NaCL per liter. It is also not physiologically normal because 0.9% saline has several features that differ from plasma. These are shown in Table 10.2. When compared to plasma, 0.9% saline has a higher sodium concentration (154 vs. 140 mEq/L), a much higher chloride concentration (154 vs. 103 mEq/L), a higher osmolality (308 vs. 290 ± 5 mOsm/L), and a lower pH (5.7 vs. 7.4). The difference in chloride concentration is most relevant, as described later.

TABLE 10.2 Crystalloid Fluids in Relation to Plasma

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Fluid	mEq/L							Omolarity (mOsm/L)
	Na	CL	K	Ca*	Mg*	Buffers	pH	
Plasma	140	103	4	2	3	HCO_3^- (25)	7.4	290
0.9% NaCL	154	154	–	–	–	–	5.7	308
Ringer's Lactate	130	109	4	3	–	Lactate (28)	6.5	273
Ringer's Acetate	131	109	4	3	–	Acetate (28)	6.7	275
Normosol®	140	98	5	–	3	Acetate (27)	7.4	295
Plasma-Lyte®						Gluconate (23)		

*Concentration of ionized calcium and total magnesium in mEq/L.
Plasma components are average or median values. Buffer concentrations in parentheses.

Volume Effects

The sodium in crystalloid fluids distributes uniformly in the extracellular fluid. Because plasma represents only 20% of the extracellular fluid, *the predominant effect of 0.9% saline infusions is to expand the interstitial volume, not the plasma volume*. This is demonstrated in Figure 10.1, which shows that infusion of one liter of 0.9% saline adds 825 mL to the interstitial volume and only 275 mL to the plasma volume (4). Note that the total increase in extracellular volume (1,100 mL) is slightly greater than the infused volume. This is due to 0.9% saline being slightly hypertonic to the extracellular fluid, which prompts a shift in water from intracellular to extracellular fluid.

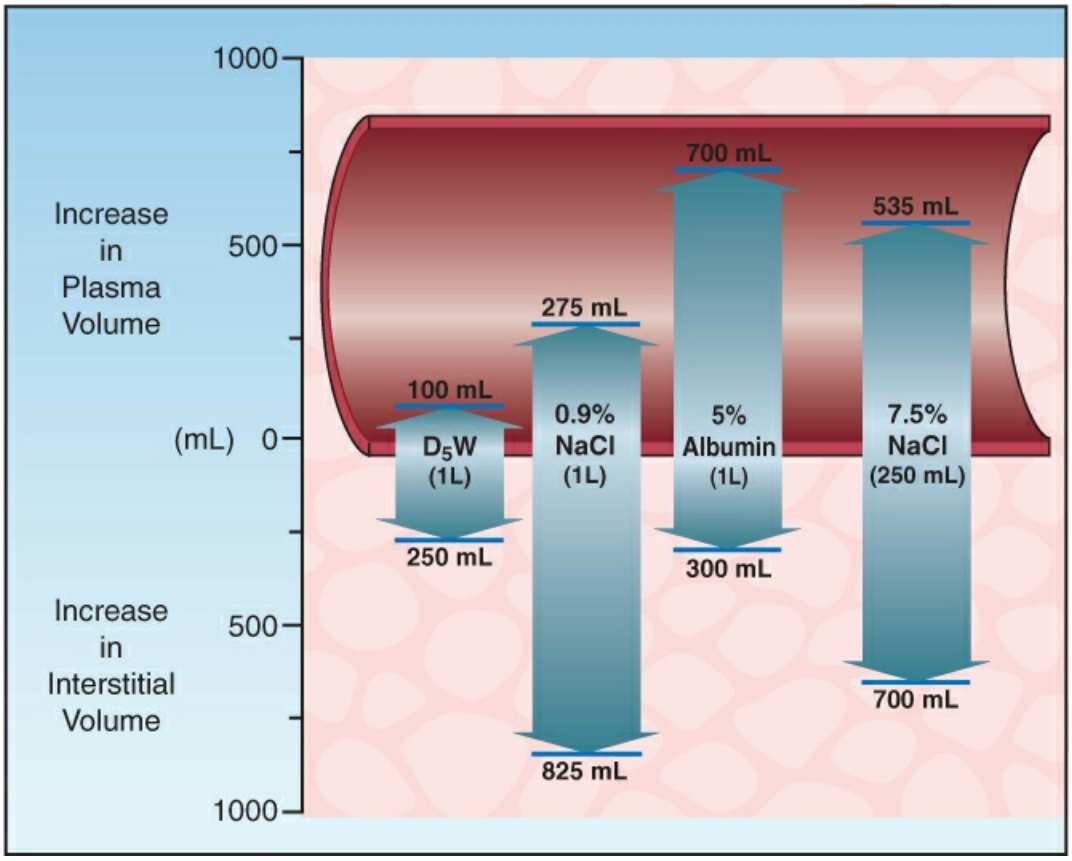


FIGURE 10.1 The effect of crystalloid and colloid fluids on expanding the plasma and interstitial fluid volumes. The infusion volume of each fluid is shown in parentheses. Data from Reference 4.

All crystalloid fluids suffer from the tendency to promote edema formation, but this tendency may be greater with 0.9% saline because of its higher sodium content (as illustrated in Figure 10.1). In addition, 0.9% saline has been implicated as a cause of acute kidney injury (see later), which will promote sodium accumulation and aggravate the tendency for edema formation.

The Evils of Chloride

There are two adverse effects of 0.9% saline that are related to the high concentration of chloride in the solution. The first of these is a *hyperchloremic metabolic acidosis*, which is the most cited adverse consequence of isotonic (0.9%) saline infusions, and usually occurs with rapid or prolonged infusions (5,6). This effect is demonstrated in Figure 10.2, which is from a study of high-volume infusions during gynecological surgery (6). The isotonic saline infusion was accompanied by a progressive decline in the blood pH (from 7.41 to 7.28), while the pH was unchanged during an equivalent infusion of Ringer's lactate solution, which has a much lower chloride concentration than isotonic saline (109 vs. 154 mEq/L), and also contains lactate as a buffer. The tendency of 0.9% saline to produce a metabolic acidosis has led to a decline in its popularity as a resuscitation fluid, especially in conditions like diabetic ketoacidosis (7).

Infusions of 0.9% saline have also been associated with *impaired renal function*, which is attributed to chloride-mediated renal vasoconstriction (5,8). Studies employing a chloride-restrictive fluid strategy (e.g., with Ringer's lactate or Plasma-Lyte) have shown a lower

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incidence of acute kidney injury (AKI) when compared with a chloride-liberal strategy using 0.9% saline (9,10). However, the AKI associated with 0.9% saline is typically mild in severity, and does not require renal replacement therapy (11).

Ringer's Lactate

The family of Ringer's solutions began with Sydney Ringer, a British physician who studied the contraction of isolated frog hearts, and who (in 1880) introduced a saline solution that contained calcium and potassium to promote cardiac contraction and cell viability (12). This solution, which is known as *Ringer's injection*, is 0.9% saline with potassium at 4 mEq/L, and calcium at 4 mg/dL.

The original Ringer's solution was modified in the early 1930's by an American pediatrician named Alexis Hartmann, who added sodium lactate as a buffer for the treatment of metabolic acidosis (12). This solution was originally called Hartmann's solution, and is now known as *Ringer's lactate* (RL) solution. The composition of this solution is shown in Table 10.2. The sodium concentration in RL is reduced to compensate for the sodium released from sodium lactate, and the chloride concentration is reduced to adjust for the lactate anion in the solution. Note that the chloride concentration in RL (109 mEq/L) is a close approximation of the chloride in plasma (103 mEq/L); as a result, RL infusions do not produce a hyperchloremic metabolic acidosis, as demonstrated in Figure 10.2.

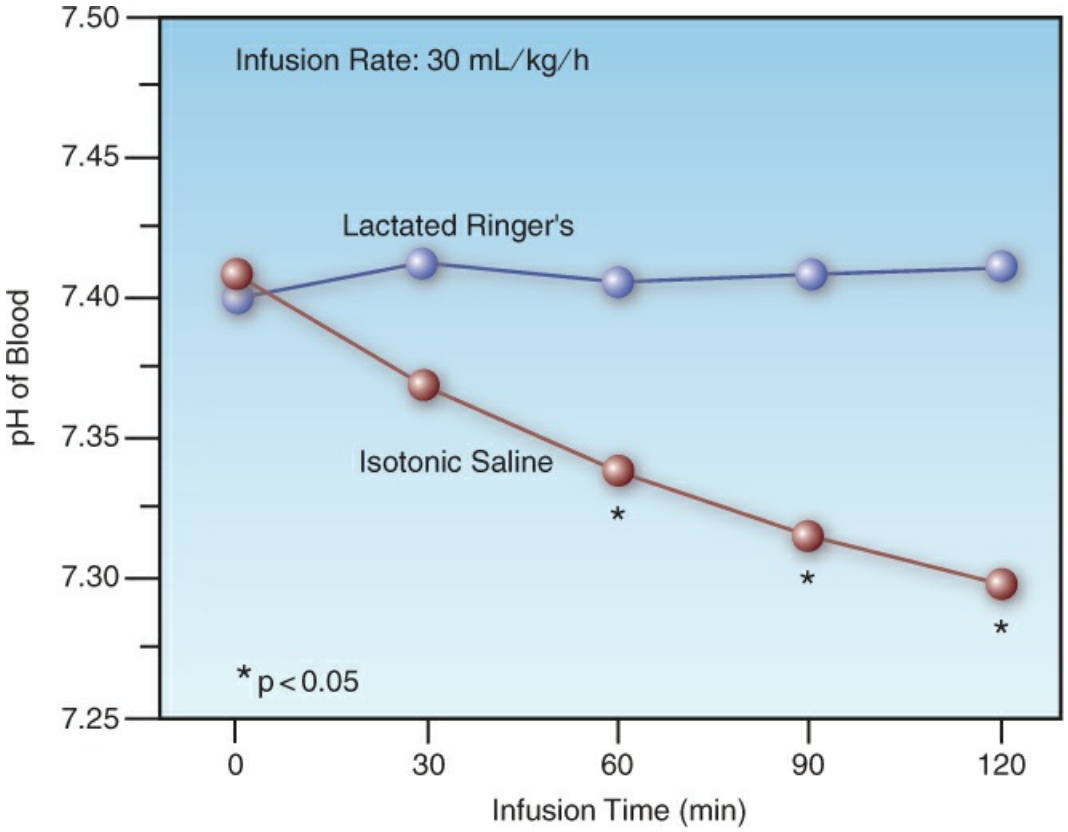


FIGURE 10.2 Cumulative effects of isotonic saline (CL = 154 mEq/L) and Ringer's lactate (CL = 109 mEq/L) on the pH of blood. Both fluids were infused at the same rate (30 mL/kg/hr). Data from Reference 6.

Lactate as a Buffer

The lactate anion in RL can serve as a buffer, but not by combining with hydrogen ions in blood. Instead, the lactate is taken up by the liver and converted to glucose, which consumes hydrogen ions (13): i.e.,

$$2\ \text{Lactate} + 2H^{+} \rightarrow \text{Glucose} \tag{10.1}$$

The hydrogen ions are donated by water, which leaves hydroxyl ions (OH-) that can combine with CO_2 to generate bicarbonate ions (HCO_3^-):

$$2\ OH^{-} + 2\ CO_2 \rightarrow 2\ HCO_3^{-} \tag{10.2}$$

Overall, each mmol of lactate generates one mmol of bicarbonate (13).

Plasma Lactate Levels

Concern about the lactate in RL (28 mmol/L) causing an increase in plasma lactate levels has received scant attention. The liver can normally clear lactate at a rate of 100 mmol/hr (14), which is equivalent to the infusion of 3.5 liters of RL per hour. In one study of healthy adult volunteers, the bolus infusion of RL at a dose of 30 mL/kg resulted in a small (0.9 mmol/L) increase in plasma lactate levels, but isotonic saline infusions had a similar effect (15). The influence of RL infusions on lactate levels in critically ill patients, who may have impaired lactate clearance from circulatory shock or hepatic insufficiency, has not been studied.

Caveat: Blood samples drawn through catheters being used for RL infusions can yield spuriously high lactate concentrations (16). Therefore, in patients receiving RL infusions, plasma lactate measurements should be obtained from sites other than the infusion catheter.

Other Issues

Other issues related to RL infusions are summarized as follows:

. The ionized calcium in RL can bind to the citrated anticoagulant in stored RBCs and promote clot formation; as a result, RL is not advised as a diluent fluid for the transfusion of erythrocyte concentrates (packed RBCs). This recommendation lacks experimental support, and there are studies showing that *rapid* infusion of RBCs and RL is safe (17). However, it still seems best to avoid RL for RBC transfusions if possible.

. The potassium in RL (4 mEq/L) is not a concern in patients with hyperkalemia, as there are no documented instances in which hyperkalemia is exacerbated by RL (18). In fact, hyperkalemia is more likely with 0.9% saline (due to the metabolic acidosis described earlier) than with RL (19).

Ringer's Acetate

Because of concerns that the lactate buffer in RL would be ineffective in patients with liver disease, and would also elevate plasma lactate levels, lactate was replaced by acetate to create *Ringer's acetate* solution (see Table 10.2), which is otherwise identical in composition to RL. Acetate is metabolized primarily in muscle (12), which makes Ringer's acetate a reasonable alternative to RL in patients with liver failure. One potential disadvantage of acetate is the

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potential for myocardial depression (5), although the clinical significance of this is unclear.

Other Balanced Salt Solutions

Normosol® and Plasma-Lyte® are balanced salt solutions that are identical in composition (see Table 10.2), and offer the following advantages:

. They have a pH that is closer to plasma pH than other crystalloid fluids.

. They contain non-lactate buffers, which eliminates the combined risks of altered buffer capacity in liver failure and spurious hyperlactatemia.

. The chloride concentration is close to that of plasma, and will not produce a hyperchloremic metabolic acidosis.

. They contain magnesium instead of calcium, and thus are safe to use with blood transfusions.

. They are more isotonic to plasma than other crystalloid fluids.

In terms of their general characteristics, these fluids are closest to the ideal plasma replacement fluid. However, their superiority in clinical settings is unproven.

Hypertonic Saline

Hypertonic saline solutions are much more effective at expanding the extracellular (plasma) volume than isotonic fluids. This is demonstrated in Figure 10.1, which shows that infusion of 250 mL of 7.5% NaCL (which has an osmolarity of 2,567 mOsm/L; 9 times that of plasma) results in a 1,235 mL increase in extracellular fluid, which is about 5 times greater than the infused volume. The added volume comes from an intracellular shift of water.

Animal studies have shown that hypertonic saline is effective for limited-volume resuscitation of hemorrhagic shock. This is demonstrated in Figure 10.3, which shows that hypertonic saline can maintain cardiac output with 1/5th the volume required with isotonic saline (20). Observations like this suggested that hypertonic saline would be well suited for situations where small volumes of resuscitation fluid are advantageous; e.g., the prehospital resuscitation of trauma victims. However, the accumulated evidence shows no apparent survival benefit from hypertonic saline compared to isotonic crystalloids for the management of traumatic shock (21).

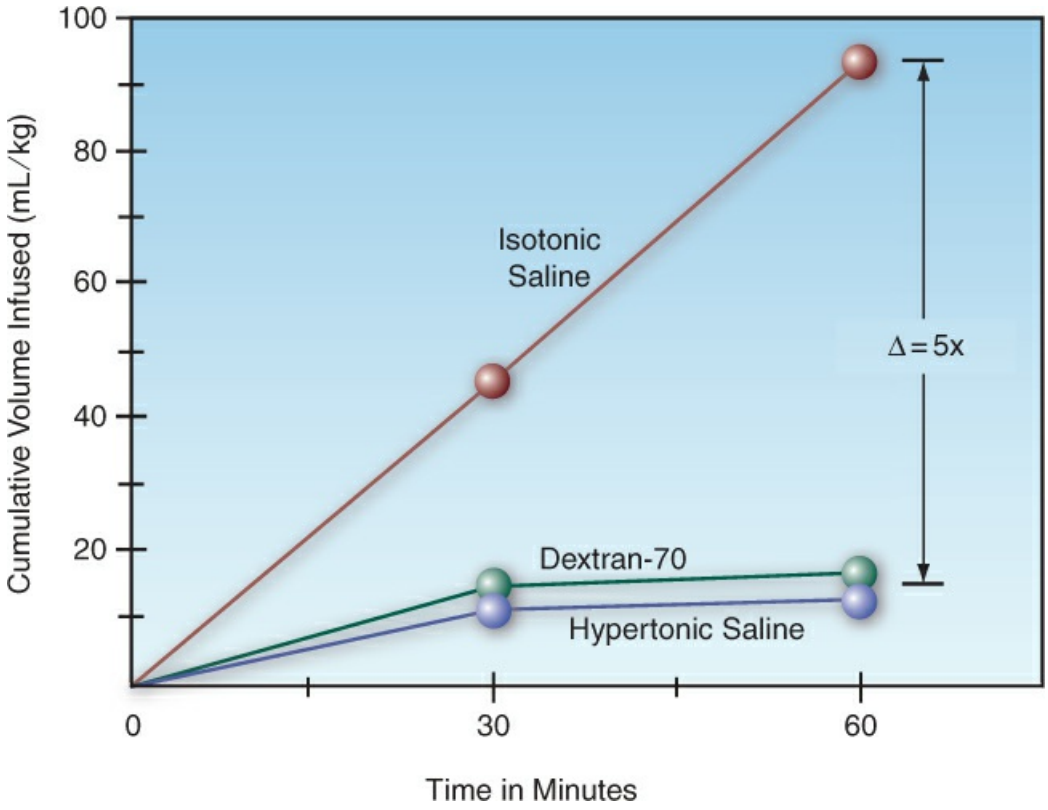


FIGURE 10.3 The cumulative volume of three intravenous fluids needed to maintain a normal rate of aortic blood flow in an animal model of hemorrhagic shock. Data from Reference 20.

Traumatic Brain Injury

The principle use of hypertonic saline is to reduce intracranial pressure (ICP) in patients with traumatic brain injury. The standard agent for this is mannitol (a 6-carbon sugar-alcohol that is also used as an osmotic diuretic), but hypertonic saline is gaining in popularity because it is equivalent to mannitol for reducing ICP, but is more likely to maintain cerebral perfusion pressure (22). Hypertonic saline also avoids the risk of acute kidney injury that has been linked to mannitol.

The saline solutions used for ICP control vary in strength from 3% to 25% saline, but 10% saline is a popular choice, and a dose of 0.6 mL/kg is equimolar to 20% mannitol at 2 mL/kg (23). Bolus doses are given every 6–8 hours as needed to keep ICP <20 mm Hg, and treatment can be continued as long as the serum osmolality is <320 mmol/kg H_2O. Unfortunately, control of ICP with hypertonic saline or mannitol has not improved the survival rate in patients with traumatic brain injury (24).

5% DEXTROSE SOLUTIONS

The once-popular use of 5% dextrose solutions (D_5 solutions) is no longer a desirable option, as explained in this section.

Protein-Sparing Effect

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Prior to the standard use of nutritional support regimens (e.g., enteral tube feedings), 5% dextrose solutions were used as a source of calories in patients who were unable to eat. Dextrose provides 3.4 kilocalories (kcal) per gram when fully metabolized, so a 5% dextrose solution (50 grams dextrose per liter) provides 170 kcal per liter. Infusion of 3 liters of a D_5 solution daily (125 mL/min) will then provide 510 kcal/day, which is enough nonprotein calories to limit the breakdown of endogenous proteins to provide calories (protein-sparing effect). This is no longer necessary, as most patients who are unable to eat will receive full nutritional support (enteral or parenteral) when needed.

Volume Effects

The addition of dextrose to intravenous fluids increases osmolarity; a 5% dextrose-in-water solution (D_5W) has an osmolarity of 278 mOsm/L, which is close to the osmolarity of plasma. However, since the dextrose is taken up by cells and metabolized, the osmotic effect rapidly wanes, and the added water then moves into cells. This explains the volume effects of D_5W shown in Figure 10.1; i.e., infusion of one liter of D_5W results in an increase in extracellular fluid (plasma plus interstitial fluid) of about 350 mL, which means the remaining 650 mL (two-thirds of the infused volume) has moved intracellularly. Thus, *the principal effect of D_5W is to expand the intracellular, not extracellular volume*, and this can result in undesirable cell swelling.

A very different effect can occur when dextrose is added to isotonic (“normal”) saline. A solution of D_5-normal saline (D_5NS) has an osmolarity of 560 mOsm/L, which is almost twice the osmolarity of plasma. Therefore, *if glucose utilization is impaired* (which is common in critically ill patients), *infusions of D_5NS can result in cellular dehydration*.

Enhanced Lactate Production

In healthy subjects, only 5% of an infused glucose load will result in lactate formation, but in patients who are hemodynamically compromised, as much as 85% of glucose metabolism can lead to lactate production (25). This latter effect is demonstrated in Figure 10.4, which is from a study where two intravenous fluids were used (Ringer’s solution and a 5% dextrose solution) to maintain cardiac filling pressures during a surgical procedure in which systemic hypoperfusion was induced by aortic cross-clamping (26). When the dextrose-containing fluid was infused, the serum lactate levels began to rise significantly after the aorta was cross-clamped, and the increase in circulating lactate levels persisted throughout the remainder of the procedure. These results indicate that, *when circulatory flow is compromised, infusion of 5% dextrose solutions can promote lactic acid production*.

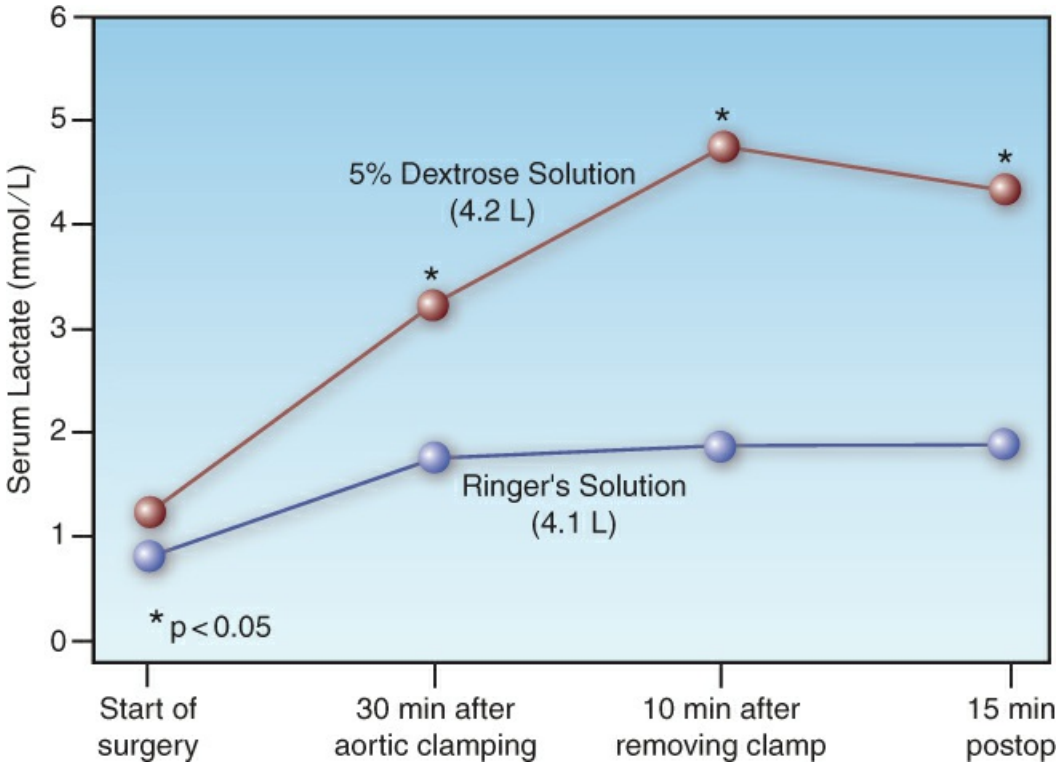


FIGURE 10.4 The effect of intravenous fluid therapy with and without dextrose on blood lactate levels in patients undergoing abdominal aortic aneurysm repair. Each point represents the mean lactate level in 10 study patients. The average infusion volume for each fluid is indicated in parentheses. Data from Reference 26.

Hyperglycemia

About 20% of patients admitted to ICUs have diabetes, and as many as 90% of patients develop hyperglycemia at some time during their ICU stay (27). Hyperglycemia has several deleterious consequences in critically ill patients, including immune suppression and increased risk of infection (28), aggravation of ischemic brain injury (29), and an increased mortality rate (27). Considering the ubiquity of hyperglycemia and the associated risks, it seems reasonable to *avoid using dextrose-containing fluids whenever possible*.

COLLOID FLUIDS

In chemical terms, a colloidal solution (also called a "suspension") is a fluid that contains insoluble particles. In clinical terms, a colloid solution is a fluid that contains large solute molecules that do not pass readily from plasma to interstitial fluid. The retained molecules create a *colloid osmotic pressure* or *oncotic pressure* (described earlier in the chapter), which holds water in the vascular compartment.

Volume Effects

The effect of volume resuscitation with a colloid fluid is demonstrated in Figure 10.1. The colloid fluid in this case is a 5% albumin solution. Infusion of one liter of this solution results in

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a 700 mL increment in the plasma volume and a 300 mL increment in the interstitial fluid volume. When compared with the increment in plasma volume after one liter of 0.9% NaCL (275 mL), the colloid fluid is about 3 times more effective in expanding the plasma volume than the crystalloid fluid. Thus, *colloid fluids will achieve a given increment in plasma volume with about 30% of the volume required with crystalloid fluids* (30).

TABLE 10.3 Colloid Fluid Comparisons

Fluid	COP (mm Hg)	Δ Plasma Volume / Infusate Volume	Duration of Effect
25% Albumin	70	3.0−4.0	12 h
10% Dextran-40	40	1.0−1.5	6 h
6% Hetastarch	30	1.0−1.3	24 h
5% Albumin	20	0.7−1.3	24 h
Plasma	28	–	–

Data from References 4,31–34. COP = colloid osmotic pressure.

Colloid Fluid Comparisons

Individual colloid fluids differ in their ability to expand the plasma volume, and this difference is a function of the colloid osmotic pressure (COP) of each fluid, and its relation to the COP of plasma (which is normally about 28 mm Hg) (31). This is demonstrated in Table 10.3, which shows the COP of common colloid fluids and the potency of each as a plasma volume expander (4,31−34). Note that fluids with higher COPs produce greater increments in plasma volume, and when the COP exceeds that of plasma, the increment in plasma volume exceeds the infusate volume. This latter point is most apparent with 25% albumin, which has a COP of 70 mm Hg (2.5 times the plasma COP), and produces an increment in plasma volume that is 3 to 4 times greater than the infusate volume.

Albumin Solutions

Albumin is responsible for about 80% of the plasma COP (31), which makes albumin solutions very desirable as plasma volume expanders. Added benefits of albumin include its role as the principal transport protein in the blood (see Table 10.4), and its antioxidant activity (35).

Characteristics

Albumin solutions are heat-treated preparations of human serum albumin that are available as a 5% solution (5 g/100 mL) and a 25% solution (25 g/mL) in 0.9% NaCL. The 5% albumin solution has a COP of 20 mm Hg, which is close to the COP of plasma. It is typically given in aliquots of 250 mL, and the plasma volume increment is at least 70% of the infused volume. The volume effect begins to dissipate at 6 hours, and is usually lost after 12 hours (4,32).

TABLE 10.4 Substances Transported by Albumin

Drugs	Others
Benzodiazepines	Bilirubin
Cephalosporins	Copper
Furosemide	Estrogen
NSAIDs	Fatty Acids
Phenytoin	Progesterone
Quinidine	Prostaglandins
Salicylates	Testosterone
Sulfonamides	Zinc
Valproic Acid	
Warfarin	

The 25% albumin solution is a hyperoncotic fluid with a COP that is 2.5 times greater than that of plasma. It is given in aliquots of 50–100 mL, and the plasma volume increment is 3 to 4 times the infusate volume. This effect is produced by fluid shifts from the interstitial space, so interstitial fluid volume decreases as plasma volume increases. Because it does not replace lost volume, but instead shifts fluid from one compartment to another, *25% albumin should not be used as a resuscitation fluid for volume loss*. This fluid is typically used to promote a rapid increase in plasma volume (and blood pressure) in patients with edema, especially when the edema is due, at least in part to hypoalbuminemia. All *albumin solutions are more effective as plasma volume expanders in the presence of hypoalbuminemia* (36).

Safety

Early reviews suggested that the use of albumin solutions had a negative impact on survival (37), but more recent studies show that albumin solutions pose no danger (38,39). One possible exception is traumatic brain injury, where one large study has shown a higher mortality associated with albumin resuscitation when compared with isotonic saline (40). Hyperoncotic (25%) albumin has been associated with an increased risk of renal impairment in patients with circulatory shock, but this risk is shared by all hyperoncotic fluids (41).

Hydroxyethyl Starches

Hydroxyethyl starch (HES) is a chemically modified polysaccharide composed of long chains of branched glucose polymers substituted periodically by hydroxyl radicals (OH), which resist enzymatic degradation. HES elimination involves hydrolysis by amylase enzymes in the bloodstream, which cleave the parent molecule until it is small enough to be cleared by the kidneys. The following is a summary of the relevant characteristics of HES preparations (34,42).

Characteristics

MOLECULAR WEIGHT: HES preparations have different molecular weights, and are classified as high MW (450 kilodaltons or kD), medium MW (200 kD) and low MW (70 kD). High MW preparations have a prolonged duration of action because amylase cleavage results in progressively smaller molecules that are osmotically active. When the cleavage products reach a molecular weight of 50 kD, they can be cleared by the kidneys (42).

MOLAR SUBSTITUTION RATIO: HES preparations are also classified by the ratio of hydroxyl radical substitutions per glucose polymer (OH/glucose), which is called the *molar substitution ratio* and ranges from zero to one (42). The molar substitution ratio identifies the type of starch in the HES preparation; e.g., 0.7 is hetastarch, 0.4 is tetrastarch. Because hydroxyl radicals resist enzymatic degradation, higher OH/glucose ratios are associated with prolonged activity. However, higher molar substitution ratios also increase the risk of adverse effects (see later).

HES PREPARATIONS: Individual HES preparations are described by their concentration, MW, and molar substitution ratio, as shown in Table 10.5. Most preparations are available as a 6% solution in 0.9% NaCL, and the molar substitution ratio is indicated in the prefix of the starch: e.g., 0.5 is pentastarch, 0.4 is tetrastarch. Hetastarch is the original HES preparation, and has a high MW(450 kD) and a high molar substitution ratio (0.7). Tetrastarch is the most recent HES preparation, and has the lowest MW (130 kD) and the lowest molar substitution ratio (0.4). Tetrastarch is available as Voluven® (6% HES 130/0.4 in 0.9% saline).

TABLE 10.5 Characteristics of Hydroxyethyl Starch Preparations

Name	Concentration	MW	MSR
Hetastarch	6%	450 kD	0.7
Hexastarch	6%	200 kD	0.6
Pentastarch	6%, 10%	200 kD	0.5
Tetrastarch	6%	130 kD	0.4

Volume Effects

The performance of 6% HES solutions as plasma volume expanders is very similar to 5% albumin. The oncotic pressure is higher than 5% albumin, and the increment in plasma volume can be higher as well (see Table 10.3). The effect on plasma volume can last up to 24 hours with high MW preparations (e.g., hetastarch) (34), while the effect with lower MW preparations (tetrastarch) can be 6 hours or even less (43).

Altered Hemostasis

HES preparations can impair hemostasis by inhibition of Factor VII and von Willebrand factor, and impaired platelet adhesiveness (42,44). The risk of altered hemostasis is highest with HES preparations that have a high molar substitution ratio (i.e., hetastarch) (42), and clinical studies have shown an increase in perioperative bleeding and transfusion requirements when hetastarch is used for volume resuscitation (45). There is evidence that tetrastarch (which has a low molar substitution ratio) does not produce a clinically significant coagulopathy until large volumes are infused (>50 mL/kg) (27).

Nephrotoxicity

Several clinical studies have shown an association between HES infusions and an increased risk of acute kidney injury (AKI) and death (46,47). However, many of these studies used an older-

generation HES preparation (e.g., hetastarch), and the patients often had circulatory shock, which increases the risk of AKI. In more recent studies using the latest generation HES preparation (tetrastarch) in less severely ill (abdominal surgery) patients, there is no association between HES infusions and AKI (48). At the present time, the risk of AKI with HES infusions is an unresolved issue (49).

Hyperamylasemia

The amylase enzymes involved in the hydrolysis of HES attach to the HES molecules, and this reduces amylase clearance by the kidneys. This can result is an increase in serum amylase levels to 2–3 times normal (34,50). Levels usually return to normal within one week after HES is discontinued. Serum lipase levels are unaffected by HES infusions (50).

The Dextrans

The dextrans are glucose polymers produced by a bacterium (*Leuconostoc*) incubated in a sucrose medium. First introduced in the 1940s, these colloids are not popular (at least in the United States) because of the perceived risk of adverse reactions. The two most common dextran preparations are 10% dextran-40 and 6% dextran-70, which have average molecular weights of 40 and 70, respectively.

Characteristics

Both dextran preparations have a COP of 40 mm Hg, and cause a greater increase in plasma volume than either 5% albumin or 6% hetastarch (see Table 10.3). Dextran-70 may be preferred because the duration of action (12 hours) is longer than that of dextran-40 (6 hours) (32).

Disadvantages

. Dextrans produce a dose-related bleeding tendency that involves impaired platelet aggregation, decreased levels of Factor VIII and von Willebrand factor, and enhanced fibrinolysis (44,50). The hemostatic defects are minimized by limiting the daily dextran dose to 20 mL/kg.

. Dextrans coat the surface of red blood cells and can interfere with the ability to cross-match blood. Red cell preparations must be washed to eliminate this problem. Dextrans also increase the erythrocyte sedimentation rate as a result of their interactions with red blood cells (50).

. Dextrans have been associated with a hyperoncotic renal injury (50,51), but this occurs rarely. Anaphylactic reactions are reported in only 0.03% of infusions (50).

THE COLLOID–CRYSTALLOID CONUNDRUM

There is a longstanding debate concerning the type of fluid that is most appropriate for volume resuscitation, and each type of fluid has its loyalists who passionately defend the merits of their chosen fluid. The following is a brief description of the issues involved in this debate.

Early Focus on Crystalloids

Early studies of acute blood loss showed that hemorrhagic shock was associated with an

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interstitial fluid deficit, partly as a result of a fluid shift from the interstitial fluid into the bloodstream (52). Studies in an animal model of hemorrhagic shock showed that replacement of the shed blood was almost universally fatal, while the mortality rate was reduced when a crystalloid fluid (Ringer's lactate) was added to the shed blood replacement (53). These results were interpreted as indicating that replacement of the interstitial fluid deficit (with a crystalloid fluid) was the critical factor in the successful resuscitation of hemorrhagic shock. This cemented the popularity of crystalloid fluids for the resuscitation of blood loss.

Current Concerns

Since those early studies, the importance of promoting cardiac output has emerged as the primary focus of volume resuscitation. To this end, colloid fluids have proven far superior to crystalloid fluids for producing acute increments in cardiac output: in one study, 500 mL of a colloid fluid (10% Dextran-40) produced a three-fold greater increment in cardiac output than one liter of a crystalloid fluid (Ringer's lactate) (54). This superiority is a reflection of the greater plasma volume expansion with colloid fluids.

Despite the enhanced ability of colloid fluids to promote cardiac output, crystalloid fluids remain the popular choice for volume resuscitation in circulatory shock. The principal argument in favor of crystalloid resuscitation is the lack of proven survival benefit with colloid resuscitation (55), and the lower cost of crystalloid fluids. The argument against crystalloid resuscitation is the relatively large volumes needed to expand the plasma volume (at least 3 times greater than the volume of colloid fluids), and the attendant risk of fluid accumulation and edema formation, which is associated with increased morbidity and mortality (56).

A Tailored Approach

The colloid-crystalloid controversy is focused on selecting one type of fluid for all clinical situations. This seems unreasonable considering the multitude of clinical scenarios encountered in critically ill patients. A more reasonable approach would be to select a fluid that is most appropriate for a specific clinical condition. The following are three clinical situations where a different resuscitation fluid would be most effective.

. In cases of life-threatening hypovolemia (where prompt restoration of the plasma volume is desirable), colloid fluid resuscitation (e.g., with 5% albumin) would be most effective.

. In cases of hypovolemia due to dehydration (where there is a uniform loss of extracellular fluid), a crystalloid fluid (e.g., Ringer's lactate) is appropriate.

. In cases of hypovolemia where hypoalbuminemia is implicated (causing fluid shifts from plasma to interstitial fluid) a hyperoncotic colloid fluid (e.g., 25% albumin) is an appropriate choice.

As demonstrated in these examples, *tailoring the type of resuscitation fluid to the specific clinical condition is a more logical approach than using the same type of fluid for all cases of hypovolemia*. This approach is "asking the right question", so it should qualify as a scientific approach.

A FINAL WORD

The following information in this chapter merits emphasis:

. Normal saline (0.9% NaCL) is not normal, chemically or physiologically, and the high chloride concentration causes a hyperchloremic metabolic acidosis. This does not occur with more balanced crystalloid fluids (e.g., Ringer's lactate).

. Isotonic crystalloid fluids expand the interstitial fluid volume more than the plasma volume. Thus, they promote fluid accumulation and edema formation, which has deleterious consequences in critically ill patients. (This topic is the focus of the next chapter).

. Dextrose-containing fluids promote lactate formation and hyperglycemia in critically ill patients, and are ill-advised for routine use.

. Colloid fluids are superior to crystalloid fluids for expanding the plasma volume and promoting cardiac output.

. Albumin solutions not only expand the plasma volume, they also support the transport function and antioxidant activity of plasma albumin.

. The hydroxyethyl starch (HES) solutions are associated with an increased risk of bleeding and renal impairment, but the newest generation of HES solutions (i.e., tetrastarch) have a lower risk of toxicity, especially when used in patients who are not critically ill (e.g., postop patients).

. The colloid-crystalloid controversy is misguided because there is no single type of fluid that is optimal for all cases of hypovolemia.

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0.9% Saline

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Colloid-Crystalloid Conundrum

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Chapter 11

Fluid Management

Restoration to normal physiology and normal function of organs … can never be achieved by inundation.

Francis Moore & G. Thomas Shires (*a*)

Humans can survive for several weeks without food, but they will succumb after just a few days without water. This need for water explains (teleologically) why the physiological response to stress or injury involves the retention of water by the kidneys (via the actions of antidiuretic hormone released from the posterior pituitary). The goal, of course, is to support cardiac output and maintain tissue perfusion. Intravenous fluid therapy has the same intent, however the infusion of fluids in critically ill patients is often excessive, especially in patients who are hemodynamically unstable, and the resulting fluid overload has a well-documented negative impact on both morbidity and mortality (1).

The frequency and deleterious consequences of iatrogenic fluid overload has led to a renewed focus on fluid management in the ICU, to identify and correct the practices that promote fluid overload (2). This chapter has a similar focus, and includes sections on the physiology of fluid resuscitation, current problems with infusion therapy, markers of tissue hypoperfusion, and the evaluation of fluid responsiveness.

VENOUS RETURN

Volume resuscitation is aimed at promoting venous return to the heart, and a misunderstanding of the physiology of venous return can result in the inappropriate use of intravenous fluids.

Determinants of Venous Return

The pressure gradient for venous return is often assumed to be the difference between the mean arterial pressure and the pressure in the right atrium, but this is not the case. An explanation for this is provided by the illustration in Figure 11.1, which shows a large sink that is filled with water, and a spigot that is adding water from an upper tank. The lower sink in this case represents the venous system (which contains 75% of the blood volume), and the upper tank represents the arterial side of the circulation. The flow of water out of the lower sink (i.e., venous

return) is driven by the hydrostatic pressure exerted by the height of the water in the sink, and not by the pressure in the upper tank (i.e., the arterial pressure). Thus, *the driving pressure for venous return is distinct from the arterial pressure* (3).

The pressure gradient that drives venous return (ΔPv) is the pressure drop from the small veins or venules to the right atrium. The upstream pressure in the small veins or venules is known as the *mean systemic pressure* (PMS), and *is equivalent to the pressure in the systemic circulation in the absence of blood flow* (4,5). Venous return can then be described as follows:

$$\text{Venous Return} = (\text{Pms} - \text{PRA})/\text{Rv} \tag{11.1}$$

where Pms is the mean systemic pressure, PRA is the right atrial pressure, and Rv is the resistance to flow in the venous circulation. Since the central venous pressure (CVP) is the clinical measure of right atrial pressure, Equation 11.1 can be rewritten as:

$$\text{Venous Return} = (\text{Pms} - \text{CVP})/\text{Rv} \tag{11.2}$$

Volume resuscitation has little impact on Rv (6), so *the goal of fluid resuscitation should be an increase in the (Pms – CVP) gradient*.

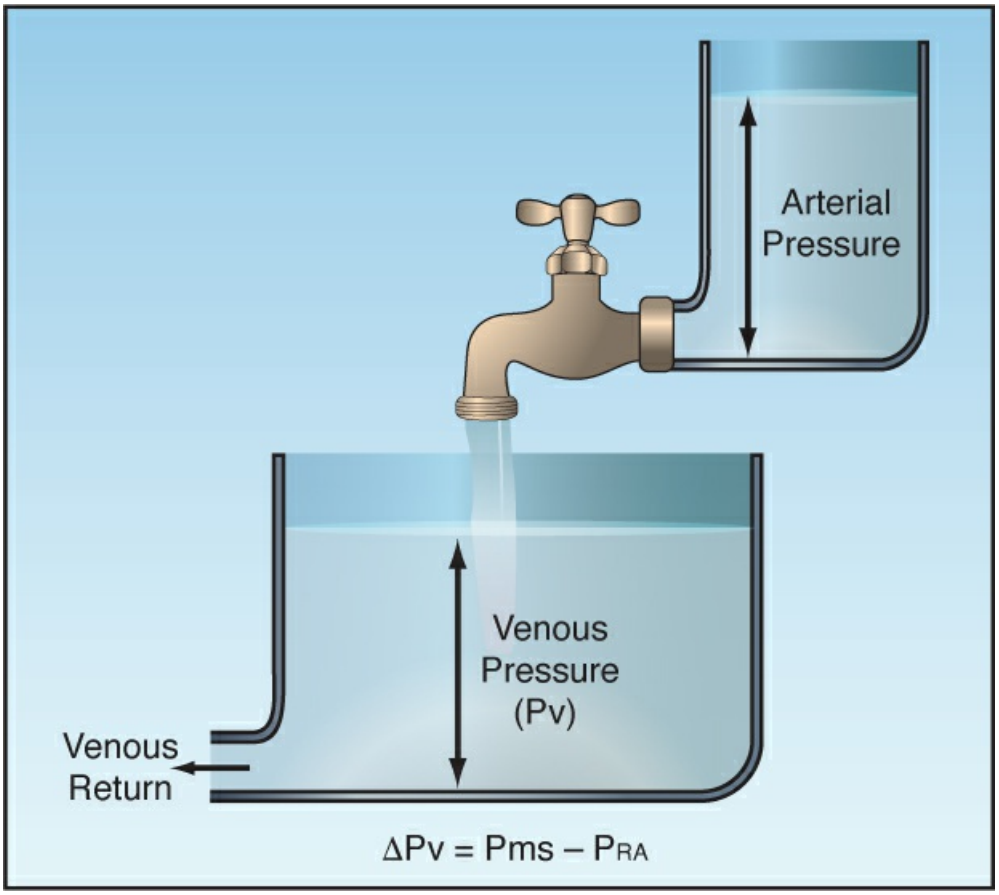


FIGURE 11.1 Illustration to demonstrate how the pressure gradient for venous return (ΔPv) is distinct from the pressure in the arterial circulation. Pms = mean systemic pressure, PRA = right atrial pressure. See text for explanation.

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Central Venous Pressure

One of the traditional goals of fluid resuscitation has been an increase in the CVP (7). However, the relationships in Equation 11.2 indicate that *an increase in the CVP (i.e., right atrial pressure) will impede venous return*. This is shown in Figure 11.2. The lines in this graph are constructed by varying the right atrial pressure (PRA) independently (i.e., without a change in blood volume) and recording the resulting changes in blood flow (venous return). Note that progressive increases in PRA above zero are associated with a steady decline in venous return: an increase in PRA of 1 mm Hg results in an average 14% decrease in venous return (8). When venous return eventually ceases, the PRA is equivalent to the Pms, and there is no pressure gradient for venous return. Note that changes in intravascular volume (i.e., hypo- and hypervolemia) are associated with the same directional change in Pms, indicating that Pms is volume-dependent.

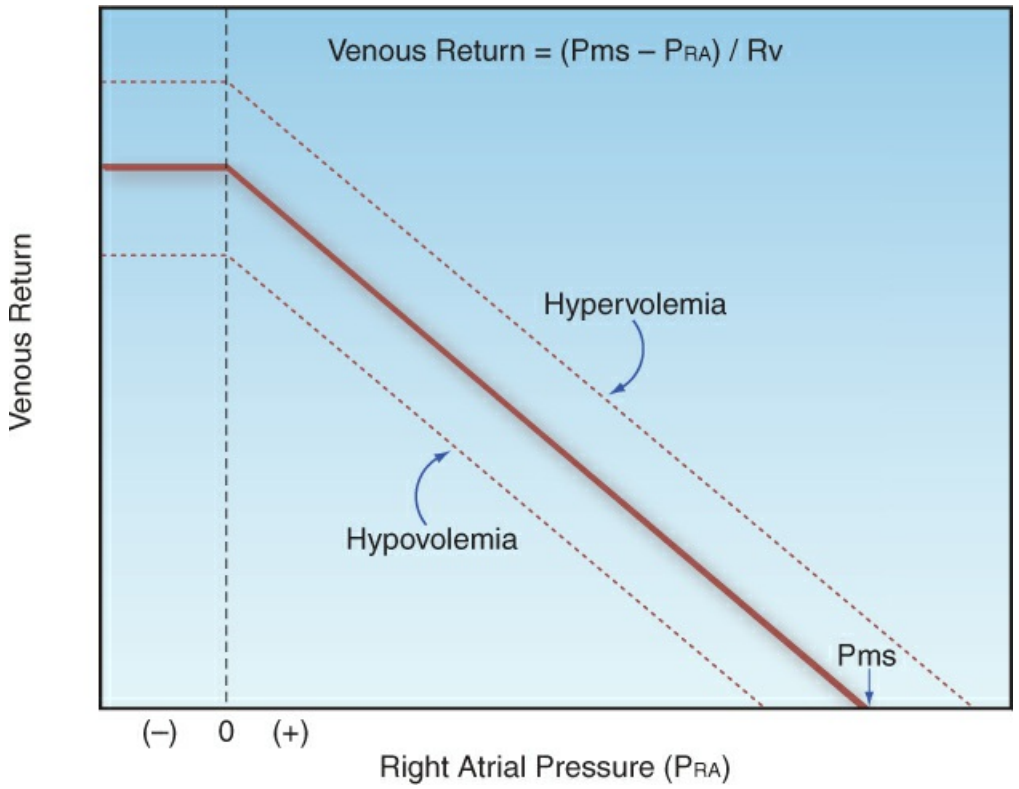


FIGURE 11.2 Graph showing the influence of right atrial pressure on venous return. The normal curve is indicated by the solid line. When venous return is zero, the right atrial pressure (PRA) is equivalent to the mean systemic pressure (Pms). See text for explanation.

Mean Systemic Pressure

The Pms is not easily measured, which may explain why there are relatively few reports of Pms measurements. The original studies involved laboratory animals, and identified a normal Pms of 7–8 mm Hg (8). Human studies have mostly involved postoperative ICU patients (some receiving vasopressors), and have reported Pms values in the 14–20 mm Hg range (9,10). The following is a brief description of two methods used to measure the Pms (9).

Stop-Flow Method

The easiest method of measuring the Pms requires an indwelling radial artery catheter and an inflatable arm cuff with a rapid cuff inflation system (D.E. Hokanson, Inc., Bellevue, WA). The cuff is placed on the upper arm and is inflated (in 0.3 sec) to a pressure that is 50 mm Hg higher than the systolic pressure recorded with the radial artery catheter. This creates a no-flow condition, and after 30 seconds of equilibration time, the radial artery pressure is equivalent to the Pms (9). The cuff is then deflated, and the measurement is repeated for a total of three measurements: the pressures should not vary by more than 5% (9).

Inflation-Hold Method

This method is restricted to ventilator-dependent patients with a regular cardiac rhythm who are also heavily sedated or not breathing spontaneously, and it requires CVP monitoring and a rapid method of measuring the cardiac output (see the section on "Fluid Responsiveness"). The basic principle is to increase the intrathoracic pressure with an "inflation-hold" maneuver (where the inflation volume is held in the lungs at end-inspiration), which will increase the CVP and decrease the cardiac output (a surrogate of venous return). A series of inflation-hold maneuvers is performed at incremental pressures (e.g., 10, 15, 20 cm H_2O) and each inflation-hold is maintained for 12 seconds. The cardiac output and CVP are recorded in the last 3 seconds of the maneuver, and the resulting values (at each pressure) are used to construct a venous return curve (like the ones in Figure 11.2); extrapolation of the curve down to the x-axis then identifies the Pms (9).

Response to Fluids

Clinical studies have consistently shown that the Pms increases in response to a fluid challenge (typically 250–500 mL) (4,5,9,10), but the increase in Pms alone does not identify a beneficial response to fluids (not if there is a similar increase in the CVP) (4). Rather, *patients who are "fluid responsive" are identified by an increase in the (Pms – CVP) gradient after a fluid challenge* (4,10), as predicted by Equation 11.2.

PROBLEMS WITH FLUID RESUSCITATION

This section highlights some of the perceptions and practices that contribute to the problem of iatrogenic fluid overload.

Plasma Volume

The target of fluid resuscitation is the plasma volume, but little consideration is given to the actual plasma volume, or the relatively small deficit that can trigger fluid resuscitation. The size of the plasma and interstitial fluid compartments are shown in Table 11.1, using estimated volumes for an average-sized adult male and female. (For the derivation of these volumes, see Table 10.1.) The total extracellular volume is 16 liters in the male, and 10 liters in the female. Plasma accounts for only 20% of the extracellular fluid, so the plasma volume is only 3 liters in the male, and 2 liters in the female. (*Note:* For a quick estimate of plasma volume, use 40 mL/kg lean body weight in males, and 36 mL/kg in females.)

TABLE 11.1 Representative Fluid Volumes in Average-Sized Adults

Fluid	Male: 75 kg	Female: 60 kg
Extracellular Fluid	16 L	10 L
Interstitial Fluid	13 L	8 L
Plasma	3 L	2 L
Clinically Significant Volume Deficit†	450 mL	300 mL

See Table 10.1 for the derivation of the fluid compartment volumes. Weights represent lean body weight. †A 15% decrease in plasma volume. From References 8 and 9.

Hypovolemia is usually clinically silent until the volume deficit reaches 15% of the blood volume (11), and this is the point that typically triggers fluid resuscitation. As shown in Table 11.1, this corresponds to a plasma volume deficit of 450 mL in the average-sized male, and 300 mL in the average-sized female. Thus, fluid resuscitation can be initiated for volume deficits of only a few hundred milliliters, and failure to consider this can lead to excessive fluid infusion.

Measuring the Plasma Volume

Although rarely used, there is a clinically validated method of measuring blood volume that uses an indicator-dilution technique with iodine-131-labeled albumin to measure plasma volume, and then calculates red cell and blood volumes based on the hematocrit (12). A specialized analyzer (BVA-100®, Daxor Corporation, Oak Ridge, TN) provides results in about one hour, which includes a comparison of the measured and normal or expected volumes. The use of this methodology in patients with septic shock provided the data shown in Figure 11.3 (13). In this case, there was a considerable increase in plasma and blood volume, which corroborates the prevalence of fluid overload in critically ill patients, and provides useful information for guiding fluid management. This approach has also been adopted for the management of acute heart failure, where measurements of plasma volume are used to guide diuretic therapy (14).

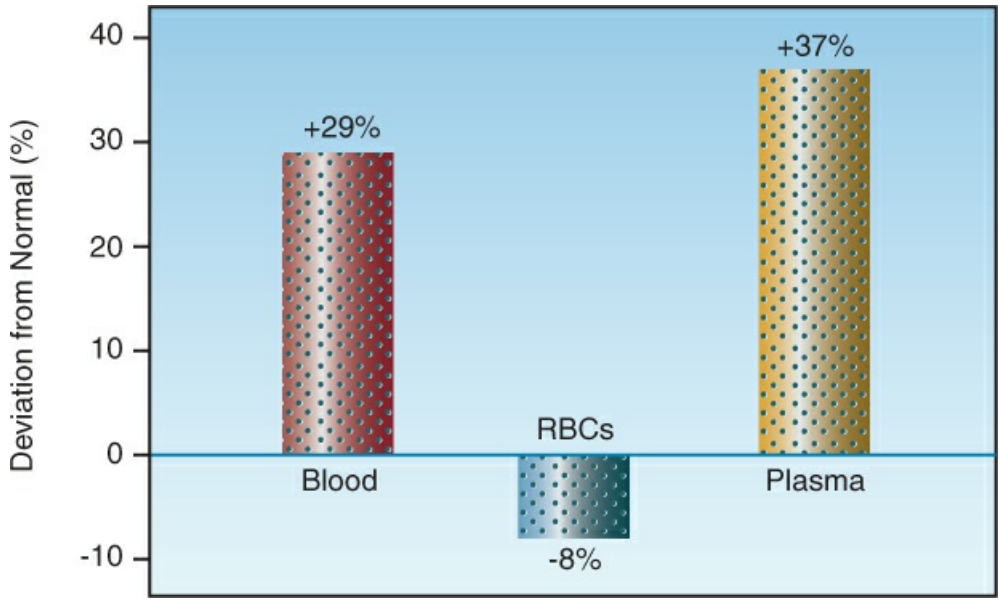


FIGURE 11.3 Measurements of plasma, red cell, and blood volume in 50 patients with septic shock, which provides evidence of iatrogenic fluid overload in critically ill patients. Height of the bars represent mean values. Data from Reference 13.

Distribution of Crystalloid Fluids

The propensity of fluid resuscitation to promote edema formation is largely due to the distribution of infused crystalloid fluids (which is described in detail in Chapter 10). Standard intravenous fluids are electrolyte (predominantly sodium chloride) solutions that distribute uniformly in the extracellular fluid, and since plasma represents 20% of the extracellular fluid (as shown in Table 11.1), then 20% of an infused crystalloid fluid will distribute in plasma, and the remaining 80% will distribute in the interstitial fluid (15). In other words, *the predominant effect of crystalloid fluids is to expand the interstitial fluid volume, not the plasma volume, and this promotes edema formation.*

EXAMPLE: The edema-forming propensity of crystalloid fluids can be demonstrated using the fluid volumes in Table 11.1. The 75 kg adult male has a plasma volume of 3 liters and a volume deficit of 450 mL (15% of the plasma volume). To replace this volume deficit with a crystalloid fluid (e.g., Ringer's lactate) will require 450/0.2 = 2.25 liters, and 1.8 liters of this fluid (2.25 × 0.8) will add to the interstitial fluid volume. Thus, replacing about half a liter of plasma has added almost two liters to the interstitial fluid.

Aggravating Factors

The tendency for crystalloid fluids to promote edema formation will be magnified by two conditions that are common in critically ill patients: i.e., hypoalbuminemia and increased capillary permeability. Albumin is responsible for 80% of the colloid osmotic pressure in plasma (the osmotic force that holds water in the intravascular compartment) (16), and thus hypoalbuminemia will promote the movement of water from plasma to interstitial fluid. Increased capillary permeability is a feature of severe inflammatory conditions, such as septic shock (described in Chapter 17) and the acute respiratory distress syndrome (described in Chapter 24). Both aggravating conditions often coexist, which adds considerably to the negative consequences of crystalloid fluid resuscitation.

Adding Colloid Fluids

Colloid fluids like 5% albumin are much more effective than crystalloid fluids for expanding the plasma volume (see Figure 10.1), and combining colloid and crystalloid fluids has been proposed for reducing the volume of infused fluids (2). This deserves attention.

Assessment of Intravascular Volume

Another major problem in fluid management is the limited ability to accurately assess the intravascular volume. The clinical evaluation of intravascular volume is so flawed that is has been called a "comedy of errors" (17). The following are some of the shortcomings.

Vital Signs

Vital signs are notoriously unreliable for the assessment of intravascular volume, especially in critically ill patients. The following are some relevant observations:

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. Tachycardia in the supine position is not observed in most cases of acute blood loss, even when the volume deficit is as high as 25% (18,19). In fact, bradycardia may be more common than tachycardia after acute blood loss (18).

. Supine hypotension does not appear until the blood volume deficit is >30% (11).

. An orthostatic increment in heart rate (≥30 bpm) is the most sensitive marker of hypovolemia, and appears when volume deficits reach 15% (18,19).

These observations indicate that tachycardia and hypotension in the supine position are insensitive markers of hypovolemia. The orthostatic increment in heart rate has limited value in the ICU, because many patients are unable to stand to perform the measurement.

Central Venous Pressure

Central venous pressure (CVP) has traditionally played a dual role in fluid resuscitation, both as a marker of intravascular volume and a target of volume resuscitation. However, neither is justified; i.e.,

. The CVP is not a reliable marker of intravascular volume. This is demonstrated by the scatterplot in Figure 11.4 where each data point represents a paired measurement of the CVP and the circulating blood volume in a group of postoperative patients (20). There is clearly no defined relationship between the two measurements, as confirmed by the correlation coefficient (r) and p value in the upper left corner of the graph. This observation has been confirmed in other studies (21).

. The CVP should not be a target of volume resuscitation, as an increase in the CVP will *impede* venous return. This is explained earlier in the chapter.

The poor correlation between the CVP and blood volume can be attributed to two confounding factors. The first is the influence of right ventricular compliance on the CVP; i.e., a decrease in ventricular compliance (distensibility) will increase the CVP. The second factor is the influence of positive intrathoracic pressure during mechanical ventilation, which is transmitted into the superior vena cava and increases the CVP. This influence can be minimized by measuring the CVP at the end of expiration (see Figure 8.5), but this does not erase the influence of positive end-expiratory pressure (PEEP).

In summary, the information just presented indicates that *the CVP should never be used to make decisions regarding fluid management*.

Inferior Vena Cava Diameter

The evaluation of intravascular volume and fluid responsiveness using respiratory variations in the diameter of the inferior vena cava (as determined by ultrasound) enjoyed a brief period of popularity, but proved to be unreliable (22).

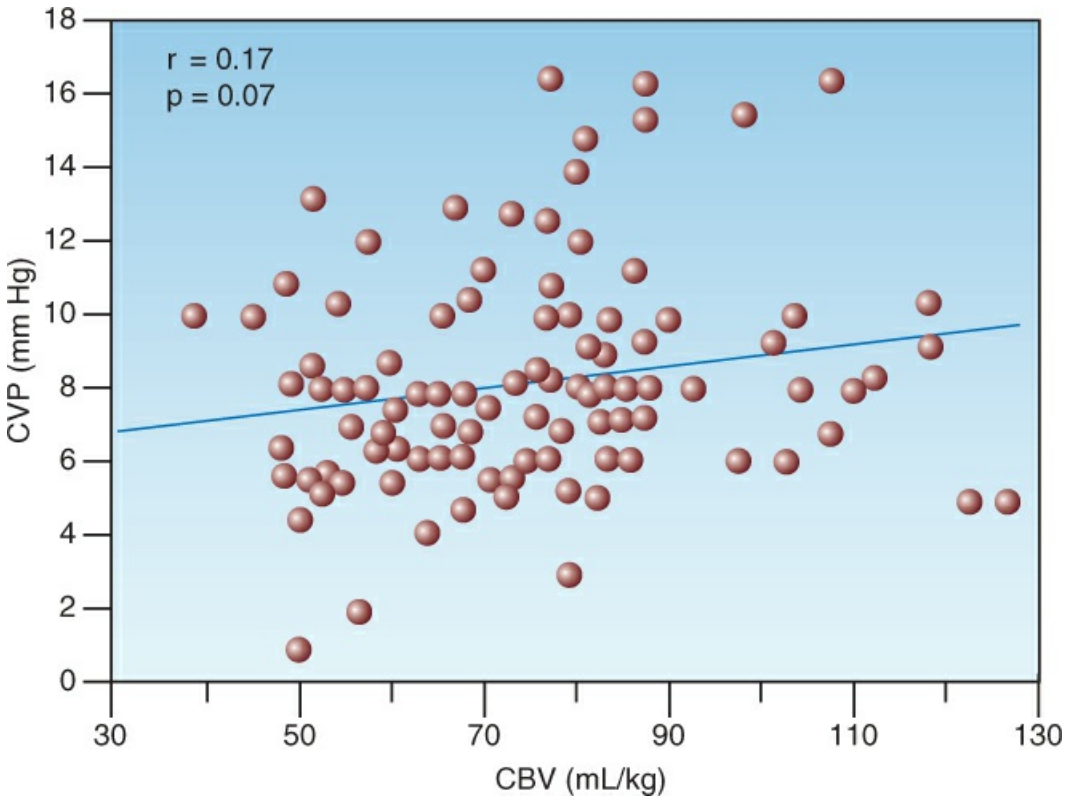


FIGURE 11.4 Scatterplot showing 112 paired measurements of circulating blood volume (CBV) and central venous pressure (CVP) in a group of postoperative patients. Correlation coefficient (R) and p value confirm the lack of correlation between the two measurements. Redrawn from Reference 20.

MARKERS OF TISSUE HYPOPERFUSION

There are two physiological markers of tissue hypoperfusion that can be used to guide fluid resuscitation: the central venous O_2 saturation ($ScvO_2$), and the venoarterial PCO_2 difference (the PCO_2 gap) (23,24). Both measurements require a central venous catheter.

Central Venous O_2 Saturation

(*Note:* This topic is described in Chapter 9, and is only briefly reviewed here.) The venous O_2 saturation (SvO_2) is a reflection of the relationship between O_2 delivery (DO_2) and O_2 consumption (VO_2). This can be explained using the following equation:

$$VO_2 = DO_2 \times (SaO_2 - SvO_2) \qquad (11.3)$$

The term ($SaO_2 - SvO_2$) expresses the extent of hemoglobin desaturation in the systemic microcirculation, also known as "O_2 extraction". The relationships in Equation 11.3 are the basis for the control of VO_2; i.e., when O_2 delivery decreases (e.g., by a decrease in cardiac output), there is a compensatory increase in O_2 extraction, and this helps to maintain a constant VO_2 (and preserve aerobic metabolism). These relationships are demonstrated in Figure 9.3. Note that as

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the DO_2 decreases, the VO_2 remains constant until the compensatory increase in ($SaO_2 - SvO_2$) reaches 50% and the SvO_2 drops to 50%. Beyond this point, the VO_2 becomes "supply dependent", which indicates the onset of aerobic metabolism. Note also that when the SaO_2 is constant, the increase in O_2 extraction is equivalent to the decrease in SvO_2.

According to the control system for VO_2 just described, *the SvO_2 will change in the same direction as changes in O_2 delivery*, and thus a decrease in SVO_2 can be used as evidence of a decrease in O_2 delivery. The SvO_2 is ideally measured in "mixed venous" blood in the pulmonary arteries, but the more accessible "central venous" O_2 saturation ($ScvO_2$) in the superior vena cava is the popular measure. The $ScvO_2$ can differ significantly from the mixed venous SvO_2 (see Chapter 9), and trends in the $ScvO_2$ are considered more reliable that individual measurements (25).

Using the $ScvO_2$

The following are some general guidelines for interpreting the $ScvO_2$, which are summarized in Table 11.2 (23). These recommendations are based on the assumption that the VO_2 is constant, and is neither abnormally high or low.

- The $ScvO_2$ is normally about 70–75%, although occasional measurements are as low as 65% or as high as 80% (23).
- An $ScvO_2$ that is <65% is evidence of a decrease in O_2 delivery. However, it is not necessarily evidence of tissue hypoperfusion, because anemia and hypoxemia are other potential causes of a decrease in O_2 delivery. (See Equation 9.8 for the determinants of O_2 delivery.) Hypoxemia is not a common culprit (because it is not allowed to persist), but anemia should be ruled out.
- A decrease in $ScvO_2$ to ≤50% indicates that tissue oxygenation may be threatened or even impaired. In this situation, prompt corrective measures (e.g., fluid resuscitation or red blood cell transfusion) are warranted, and the plasma lactate level should be measured.
- The $ScvO_2$ can be higher than expected in patients with septic shock, and may reach levels above 80%. This is commonly attributed to peripheral shunting from high flow rates, but the most likely source is a defect in O_2 utilization in tissues. (This is described in more detail in Chapter 17.) In this situation, the PCO_2 gap can help to uncover tissue hypoperfusion.

ANEMIA: A decrease in $ScvO_2$ can be a consequence of anemia; an $ScvO_2$ <70% has been proposed as an indication for red blood cell transfusions (26). When the $ScvO_2$ is <65% and anemia is present, the PCO_2 gap can help to determine if hypoperfusion is present.

TABLE 11.2 Markers of Tissue Hypoperfusion

	Normal Range	Tissue Hypoperfusion	Tissue O_2 Threatened
$ScvO_2$	65–75%	<65%	≤50%
PCO_2 Gap	2–5 mm Hg (0.3–0.7 kPa)	>6 mm Hg (>0.8 kPa)	—

The PCO_2 Gap

The PCO_2 gap refers to the difference in PCO_2 between venous and arterial blood, which is a function of blood flow. This can be explained using the Fick principle to derive the rate of CO_2 production (VCO_2); i.e.,

$$VCO_2 = CO \times (CvCO_2 - CaCO_2) \quad (11.4)$$

where CO is cardiac output and ($CvCO_2 - CaCO_2$) is the difference in CO_2 content between venous and arterial blood. The CO_2 content can be expressed as the product of the PCO_2 and a dissociation constant (k), so Equation 11.3 can be rewritten as (23):

$$VCO_2 = CO \times k \times (PvCO_2 - PaCO_2) \quad (11.5)$$

Rearranging the terms then yields the following relationships:

$$(PvCO_2 - PaCO_2) = VCO_2 / CO \times k \quad (11.6)$$

This equation reveals that *the venoarterial PCO_2 difference (the PCO_2 gap), is inversely related to the cardiac output*, and thus a decrease in cardiac output will increase the PCO_2 gap (27).

The PCO_2 gap is thus a reflection of the CO_2 washout from tissues. Carbon dioxide is a major product of metabolism, and must be continually removed in venous blood. Impaired tissue perfusion leads to a buildup of CO_2 in tissues and, because CO_2 is highly diffusible, this results in an increase in the venous PCO_2 (and an increase in the PCO_2 gap).

Using the PCO_2 Gap

The PCO_2 gap is a specific marker of tissue hypoperfusion, and it has been recommended as a target of fluid resuscitation in patients with circulatory shock, including septic shock (23,24,27,28). The following is a summary of the relevant points about the PCO_2 gap.

- The venous PCO_2 in the superior vena cava is used for the PCO_2 gap.
- The normal PCO_2 gap is 2–5 mm Hg (0.3 – 0.7 kPa).
- A PCO_2 gap that is >6 mm Hg (>0.8 kPa) is evidence of tissue hypoperfusion (27).
- The PCO_2 gap provides no information about the adequacy of tissue oxygenation, and a normal PCO_2 gap does not rule out impaired tissue oxygenation.
- The Haldane effect (i.e., hypoxia increases CO_2 binding to hemoglobin) alters the relationship between the PCO_2 and CO_2 content in venous blood, and this can affect the reliability of the PCO_2 gap. However, the clinical significance of this is unclear.

Comment

The PCO_2 gap has two advantages over the $ScvO_2$: 1) It is more specific for detecting tissue hypoperfusion, and 2) the reliability is not affected by septic shock. Because of these advantages,

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the PCO_2 gap should be preferred for monitoring tissue perfusion, especially in patients with septic shock or anemia. However, because the PCO_2 gap cannot detect impaired tissue oxygenation, combining both measurements may be the optimal approach in critically ill patients.

FLUID RESPONSIVENESS

Aggressive fluid resuscitation is common in hemodynamically unstable patients, but *only about 50% of hemodynamically unstable patients show a beneficial response to fluids* (i.e., an increase in cardiac stroke output) (29). Therefore, evaluating patients for fluid responsiveness should have a significant impact on iatrogenic fluid overload. It is important to emphasize that *fluid responsiveness does not imply hypovolemia*, since patients who are normovolemic or even hypervolemic can be fluid responsive.

The Fluid Challenge

The standard method for determining fluid responsiveness is to administer a fluid bolus, which is usually *500 mL of a crystalloid fluid infused over 10–15 minutes* (30,31). The rate of infusion is more important than the volume infused (30).

The Response

A positive response to a fluid challenge is defined as a >10% increase in stroke volume (31). This response peaks within minutes after the bolus, and dissipates after 10 minutes (32), which is too evanescent to be measured by the thermodilution technique using pulmonary artery catheters. Two methods of continuous cardiac output monitoring are favored in this situation, as described next.

PULSE CONTOUR ANALYSIS: The arterial pressure waveform can be used to measure the stroke volume, as illustrated in the upper panel of Figure 11.5 (33). The shaded "area under the curve" (AUC) represents the systolic portion of the waveform. The stroke volume is directly related to the AUC, and is inversely related to the cardiac afterload (i.e., aortic compliance and resistance). This method requires an arterial catheter, a suitable arterial pressure waveform, and a regular cardiac rhythm. Measurements obtained with a commercially available monitoring system (FloTrac® System, Edwards Laboratories, Irvine, CA) are accurate in most patients, but can slightly underestimate the stroke volume in conditions of low vascular resistance (e.g., septic shock) (34). However, the system is considered reliable for evaluating fluid responsiveness (34).

ESOPHAGEAL DOPPLER METHOD: Continuous cardiac output monitoring is also possible with a Doppler probe placed in the distal esophagus to measure flow velocity in the descending thoracic aorta. The area under the aortic velocity wave is then multiplied by the aortic cross-sectional area to derive the stroke volume (35). Measurements obtained with this method compare favorably with the thermodilution method of measuring cardiac output (the gold standard) (36), but aortic stenosis produces spurious recordings (35).

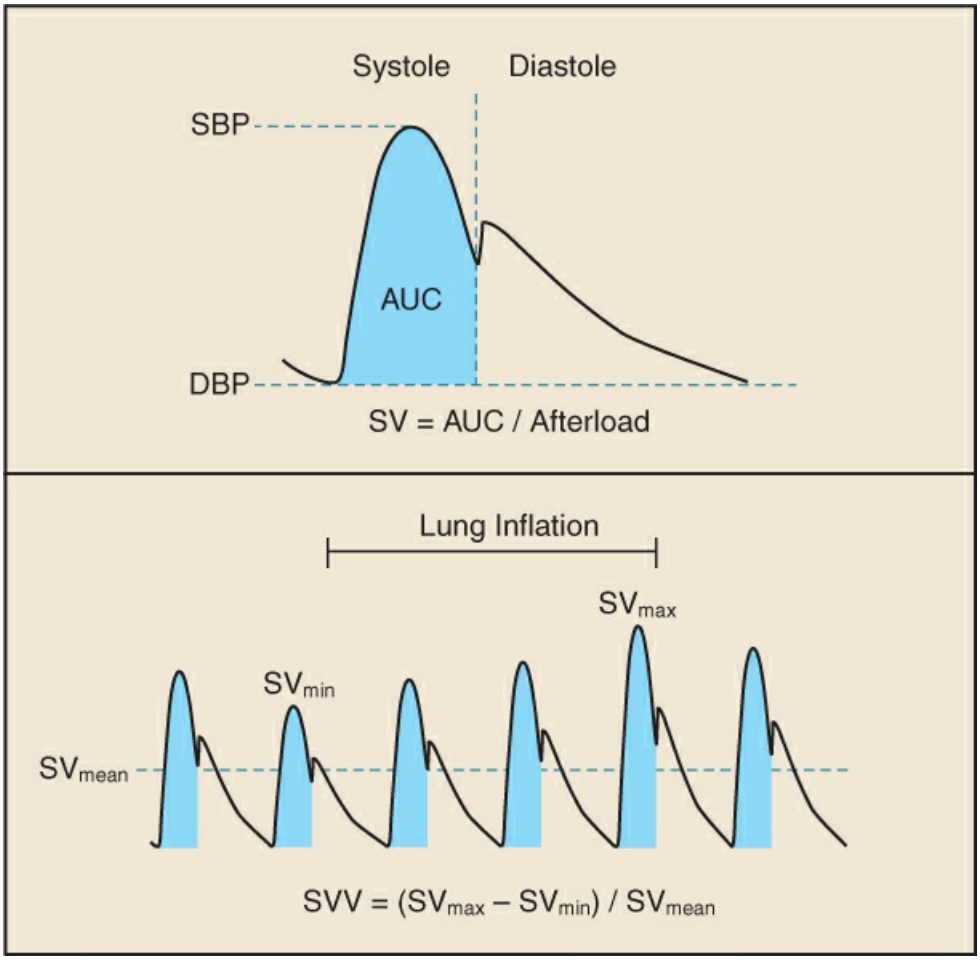


FIGURE 11.5 The upper panel shows the pulse contour method of measuring the stroke volume (SV), where the area under the curve (AUC) represents the systolic portion of the arterial pressure waveform. The lower panel demonstrates the respiratory variation in stroke volume that occurs during mechanical ventilation, and shows how this stroke volume variation (SVV) is quantified. See text for further explanation.

No Fluid Challenge

The use of fluid boluses to evaluate fluid responsiveness can be counterproductive by adding to the problem of fluid overload. This is not a concern with the following "fluid-free" methods of evaluating fluid responsiveness.

Stroke Volume Variation

During controlled mechanical ventilation, the positive-pressure lung inflation is accompanied by an increase in cardiac stroke output, as illustrated in lower panel of Figure 11.5. This is explained by the ability of positive intrathoracic pressure to decrease left ventricular afterload (see Chapter 26), and this effect produces cyclical changes in stroke volume during mechanical ventilation. This stroke volume variation (SVV) can be quantified as shown in Figure 11.5, and clinical studies have shown that an SVV of ≥15% can identify fluid responsiveness in 80% of cases (37). SVV monitoring is restricted to patients who have no breathing efforts and no arrhythmias, and this limits its use.

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Passive Leg Raising

Elevating the legs to 45° above the horizontal plane will move about 300 mL of blood out of the lower extremities and towards the heart (38), thereby serving as a “built-in” fluid challenge. This passive leg raising (PLR) maneuver has been shown to promote venous return by increasing mean systemic pressure (39). The subsequent increase in stroke volume is rapid, and can dissipate in minutes (38). A ≥10% increase in stroke volume after PLR has a sensitivity of 85% and a specificity of 92% for identifying a positive response to a fluid bolus (40). False negative results are common in patients with increased intraabdominal pressure (41).

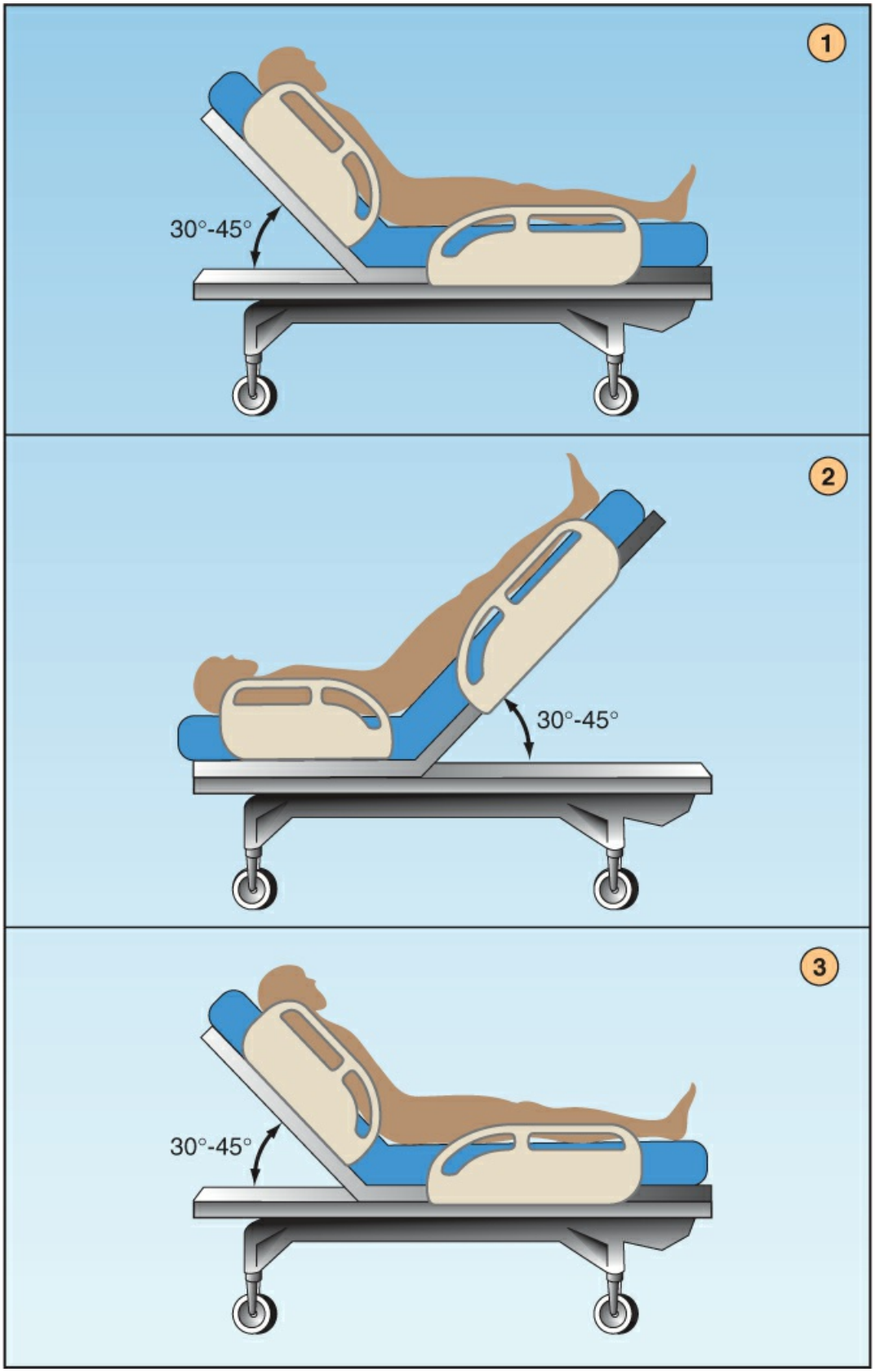


FIGURE 11.6 The passive leg raising test for fluid responsiveness. See text for explanation.

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The PLR test is performed on spontaneously breathing patients (on or off the ventilator), and the response is evaluated using a method of continuous cardiac output monitoring (e.g., pulse contour analysis). Blood pressure monitoring alone is not adequate for evaluating the response (38). The test is illustrated in Figure 11.6, and should proceed as follows (using the numbers that correspond to each panel in the figure):

. The patient should be in a semirecumbent position (head up at 30–45°) at the beginning of the test, since starting from the supine position will eliminate the contribution of blood mobilized from the splanchnic circulation (38).
. After baseline measurements of stroke volume, the hospital bed is used to move the patient into a position where the trunk is horizontal and the legs are elevated to 30–45°. (The bed is used for this purpose to reduce pain from shifting positions, which could produce a false positive result.)
. If there is a positive response (i.e., a ≥10% increase in stroke volume) that persists after a few minutes, the patient should be moved back to a semirecumbent position, and the response should dissipate.

As mentioned, pain triggered by the change in position can produce a positive response to PLR, and this should be suspected if the response is accompanied by tachycardia (since PLR should not cause tachycardia) (38), or if the response persists after returning the patient to a semirecumbent position.

Comment

The need for continuous cardiac output monitoring to evaluate fluid responsiveness creates several drawbacks, including costly equipment, the need for trained personnel (for esophageal Doppler monitoring) or a radial artery catheter (for pulse contour analysis), plus a regular cardiac rhythm, and (for monitoring stroke volume variation) patients with little or no spontaneous breathing efforts. Noninvasive methods (e.g., respiratory variations in pulse pressure, changes in end-tidal PCO_2 after passive leg raising) have been evaluated, but none have proven reliable enough to replace existing methods.

Considerations

The goal of determining fluid responsiveness is to reduce the risk of iatrogenic fluid overload, but it seems that this can be achieved with less costly and convoluted methods, such as the ones listed below.

. When aggressive fluid resuscitation is indicated (e.g., for the initial management of hypotension), using a combination of colloid and crystalloid fluids instead of crystalloid fluids alone will reduce the volume requirement.
. If there is no clinical response to the initial period of fluid resuscitation, then prompt de-escalation of the fluid regimen is indicated.
. Avoid daily maintenance fluids if there is adequate oral or enteral fluid intake.
. Pay attention to daily fluid intake and output, and correct positive daily fluid balances

promptly (with a diuretic if needed). This will prevent “fluid creep”, which is a common and often overlooked source of iatrogenic fluid overload (42).

A FINAL WORD

Getting Rid of Fluid

The human body responds to fluid deprivation by conserving salt and water, which is a robust protective response with multiple components (e.g., activation of the renin-angiotensin-aldosterone system, release of ADH). On the other end of the spectrum, our ability to protect against fluid overload seems feeble by comparison. For example, it can take over 2 days to excrete a 2-liter infusion of isotonic saline (43). This discrepancy could be a reflection of fluid control systems that operate at a low baseline level, where activating the system will have a far greater impact than inhibiting it. The sluggish response to fluids could also be a distribution problem, since most of the infused volume of crystalloid fluids is distributed outside the vascular space (where it is unlikely to trigger any type of compensatory response). Whatever the reason, our apparent meager ability to deal with fluid excess emphasizes the value of a conservative approach to intravenous fluid therapy.

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